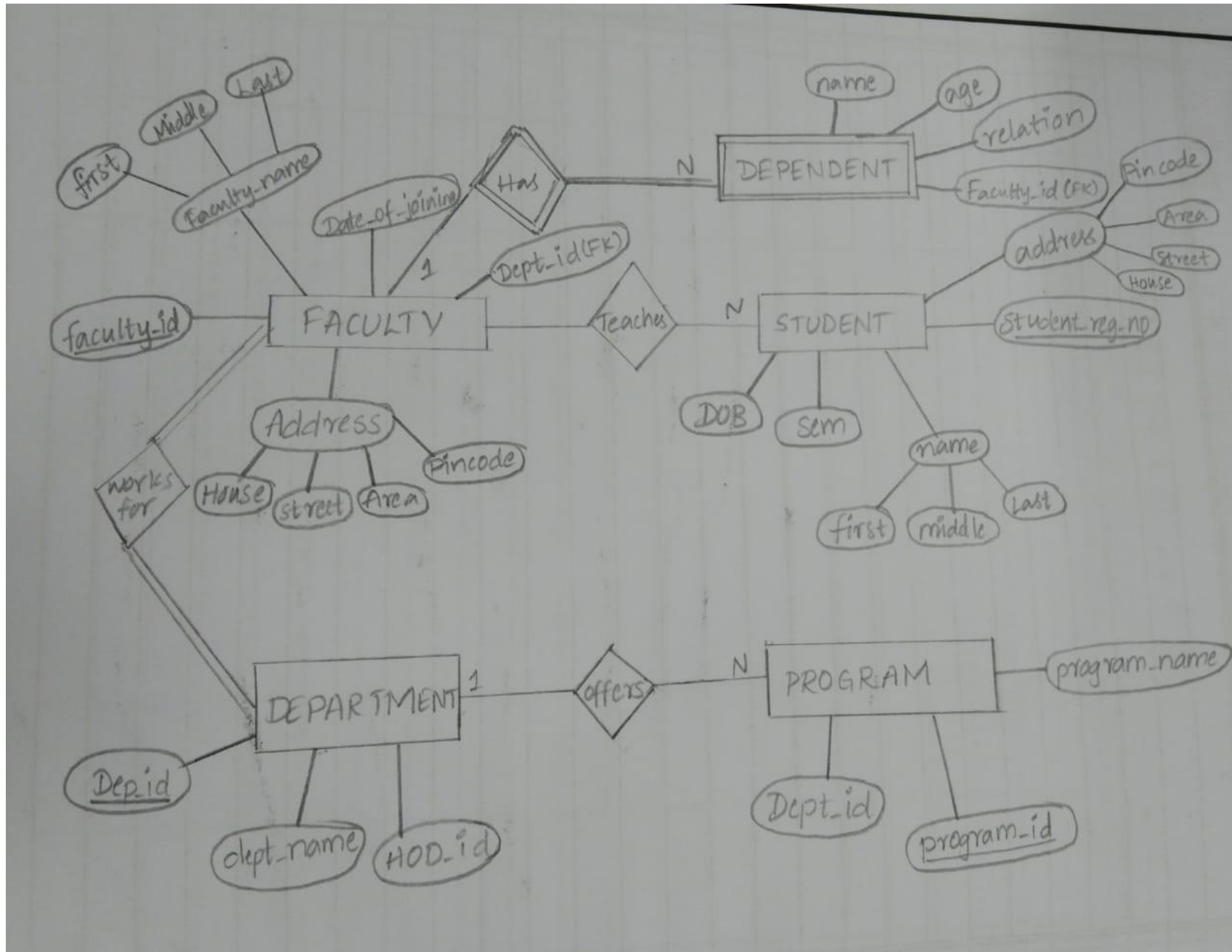


ER diagram



The University Database has the following tables:

STUDENT(usn, name, address, date_of_birth, program_Id, semester)

DEPARTMENT(dept_Id, dept_name, HOD_Id) (HOD_Id is foreign key which references faculty_Id of FACULTY table) FACULTY(faculty_Id, name, address, date_of_join, dept_Id)

PROGRAM (program_Id, program_name, dept_Id) (Each Department offers multipleprogrammes eg: BCA, MCA are offered by CS department)

DEPENDENT(faculty_id,name,phone number,age,relation)

```
CREATE TABLE FACULTY_22PHCS004A (  
    faculty_id VARCHAR(10) PRIMARY KEY,  
    faculty_name VARCHAR(30),  
    faculty_address VARCHAR(200),  
    faculty_date_of_joining DATE,  
    dept_id INT  
);
```

Table created.

```
describe faculty_22PHCS004A;
```

entity [faculty_22PHCS004A]

Name [ATTRIBUTE]	Null?	Type

FACULTY_ID	NOT NULL	VARCHAR2(10)
FACULTY_NAME		VARCHAR2(30)
FACULTY_ADDRESS		VARCHAR2(200)
FACULTY_DATE_OF_JOINING		DATE
DEPT_ID		NUMBER(38)
SALARY		NUMBER(10,2)

```
CREATE TABLE DEPARTMENT_22PHCS004A (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(100),  
    HOD_id VARCHAR(10),  
    FOREIGN KEY (HOD_id) REFERENCES FACULTY_22PHCS004A(faculty_id)  
);
```

Table created.

describe department_22PHCS004A;

Name	Null?	Type

DEPT_ID	NOT NULL	NUMBER(38)

DEPT_NAME VARCHAR2(100)
HOD_ID VARCHAR2(10)

```
CREATE TABLE PROGRAM_22PHCS004A (  
    program_id varchar(10) PRIMARY KEY,  
    program_name VARCHAR(100),  
    dept_id INT,  
    FOREIGN KEY (dept_id) REFERENCES DEPARTMENT_22PHCS004A(dept_id)  
);
```

Table created.

```
describe program_22PHCS004A;
```

Name	Null?	Type

PROGRAM_ID	NOT NULL	VARCHAR2(10)
PROGRAM_NAME		VARCHAR2(100)
DEPT_ID		NUMBER(38)

```
CREATE TABLE STUDENT_22PHCS004A (  
    student_reg_no VARCHAR(20) PRIMARY KEY,  
    student_name VARCHAR(30),  
    student_address VARCHAR(200),
```

```
student_date_of_birth DATE,  
program_id varchar(10),  
semester INT,  
FOREIGN KEY (program_id) REFERENCES PROGRAM_22PHCS004A(program_id)  
);
```

Table created.

```
describe student_22PHCS004A;
```

Name	Null?	Type

STUDENT_REG_NO		NOT NULL VARCHAR2(20)
STUDENT_NAME		VARCHAR2(30)
STUDENT_ADDRESS		VARCHAR2(200)
STUDENT_DATE_OF_BIRTH		DATE
PROGRAM_ID		VARCHAR2(10)
SEMESTER		NUMBER(38)

```
INSERT INTO FACULTY_22PHCS004A (faculty_id, faculty_name, faculty_address, faculty_date_of_joining, dept_id) VALUES (1, 'Kushi', '123 Main St', TO_DATE('2020-01-15', 'YYYY-MM-DD'), 10);  
  
INSERT INTO FACULTY_22PHCS004A (faculty_id, faculty_name, faculty_address, faculty_date_of_joining, dept_id) VALUES (2, 'Alya', '234 Main St', TO_DATE('2019-08-10', 'YYYY-MM-DD'), 20);  
  
INSERT INTO FACULTY_22PHCS004A (faculty_id, faculty_name, faculty_address, faculty_date_of_joining, dept_id) VALUES (3, 'Arav', '345 Main St', TO_DATE('2018-05-22', 'YYYY-MM-DD'), 30);
```

```
INSERT INTO FACULTY_22PHCS004A (faculty_id, faculty_name, faculty_address, faculty_date_of_joining, dept_id) VALUES (4, 'Jo', '456 Main St', TO_DATE('2021-02-18', 'YYYY-MM-DD'), 40);

INSERT INTO FACULTY_22PHCS004A (faculty_id, faculty_name, faculty_address, faculty_date_of_joining, dept_id) VALUES (5, 'Anav', '567 Main St', TO_DATE('2020-01-15', 'YYYY-MM-DD'), 50);
```

```
set linesize 150

COLUMN faculty_id FORMAT A10
COLUMN faculty_name FORMAT A20
COLUMN faculty_address FORMAT A30
COLUMN faculty_date_of_joining FORMAT A25
COLUMN dept_id FORMAT 99999

select * from FACULTY_22PHCS004A;
```

OUTPUT: TABLE-1

TABLE-1

FACULTY_ID	FACULTY_NAME	FACULTY_ADDRESS	FACULTY_DATE_OF_JOINING	DEPT_ID
1	Kushi	123 Main St	15-JAN-20	10
2	Alya	234 Main St	10-AUG-19	20
3	Arav	345 Main St	22-MAY-18	30
4	Jo	456 Main St	18-FEB-21	40
5	Anav	567 Main St	15-JAN-20	50

FACULTY_ID	FACULTY_NAME	FACULTY_ADDRESS	FACULTY_DATE_OF_JOINING	DEPT_ID
1	Kushi	123 Main St	15-JAN-20	10
2	Alya	234 Main St	10-AUG-19	20
3	Arav	345 Main St	22-MAY-18	30
4	Jo	456 Main St	18-FEB-21	40
5	Anav	567 Main St	15-JAN-20	50

```
INSERT INTO DEPARTMENT_22PHCS004A (dept_id, dept_name, HOD_id) VALUES (10, 'Computer Science', 1);
```

```
INSERT INTO DEPARTMENT_22PHCS004A (dept_id, dept_name, HOD_id) VALUES (20, 'Chemistry', 2);
```

```
INSERT INTO DEPARTMENT_22PHCS004A (dept_id, dept_name, HOD_id) VALUES (30, 'English', 3);
```

```
INSERT INTO DEPARTMENT_22PHCS004A (dept_id, dept_name, HOD_id) VALUES (40, 'Physics', 4);
```

```
INSERT INTO DEPARTMENT_22PHCS004A (dept_id, dept_name, HOD_id) VALUES (50, 'Maths', 5);
```

```
COLUMN dept_id FORMAT 99999
```

```
COLUMN dept_name FORMAT A30
```

```
COLUMN HOD_id FORMAT A10
```

```
SELECT * FROM DEPARTMENT_22PHCS004A;
```

OUTPUT: TABLE-2

TABLE-2

DEPT_ID	DEPT_NAME	HOD_ID
10	Computer Science	1
20	Chemistry	2
30	English	3
40	Physics	4
50	Maths	5

DEPT_ID	DEPT_NAME	HOD_ID
-----	-----	
10	Computer Science	1
20	Chemistry	2
30	English	3
40	Physics	4
50	Maths	5

```
INSERT INTO PROGRAM_22PHCS004A (program_id, program_name, dept_id) VALUES (101, 'BCA', 10);
INSERT INTO PROGRAM_22PHCS004A (program_id, program_name, dept_id) VALUES (102, 'BCA', 20);
INSERT INTO PROGRAM_22PHCS004A (program_id, program_name, dept_id) VALUES (103, 'BA English', 30);
INSERT INTO PROGRAM_22PHCS004A (program_id, program_name, dept_id) VALUES (104, 'BSC Physics', 40);
INSERT INTO PROGRAM_22PHCS004A (program_id, program_name, dept_id) VALUES (105, 'BSC Maths', 50);
```

```
COLUMN program_id FORMAT A10
COLUMN program_name FORMAT A20
COLUMN dept_id FORMAT 99999
SELECT * FROM PROGRAM_22PHCS004A;
```

OUTPUT: TABLE-3

TABLE-3

PROGRAM_ID	PROGRAM_ID PROGRAM_NAME DEPT_IDs	PROGRAM_ID PROGRAM_NAME DEPT_IDs
101	BCA	10
102	BCA	20
103	BA English	30
104	BSC Physics	40
105	BSC Maths	50

PROGRAM_ID PROGRAM_NAME DEPT_IDs

101 BCA 10

102	BCA	20
103	BA English	30
104	BSC Physics	40
105	BSC Maths	50

```
INSERT INTO STUDENT_22PHCS004A (student_reg_no, student_name, student_address, student_date_of_birth, program_id, semester)
VALUES ('S001', 'Madesh', '123 main st', TO_DATE('2004-08-26', 'YYYY-MM-DD'), 101, 5);
```

```
INSERT INTO STUDENT_22PHCS004A (student_reg_no, student_name, student_address, student_date_of_birth, program_id, semester)
VALUES ('S002', 'Suma', '456 Walnut Ave', TO_DATE('1999-06-05', 'YYYY-MM-DD'), 102, 5);
```

```
INSERT INTO STUDENT_22PHCS004A (student_reg_no, student_name, student_address, student_date_of_birth, program_id, semester)
VALUES ('S003', 'Keerthi', '789 Elm St', TO_DATE('2021-06-01', 'YYYY-MM-DD'), 103, 4);
```

```
INSERT INTO STUDENT_22PHCS004A (student_reg_no, student_name, student_address, student_date_of_birth, program_id, semester)
VALUES ('S004', 'Vaishu', '101 Cedar Rd', TO_DATE('2003-09-10', 'YYYY-MM-DD'), 104, 3);
```

```
INSERT INTO STUDENT_22PHCS004A (student_reg_no, student_name, student_address, student_date_of_birth, program_id, semester)
VALUES ('S005', 'Pooja', '202 Ash Blvd', TO_DATE('2003-10-07', 'YYYY-MM-DD'), 105, 2);
```

```
COLUMN student_reg_no FORMAT A15
```

```
COLUMN student_name FORMAT A20
```

```
COLUMN student_address FORMAT A30
```

```
COLUMN student_date_of_birth FORMAT A25
```

```
COLUMN program_cd FORMAT A10
```

```
COLUMN semester FORMAT 99999
```

```
SELECT * FROM STUDENT_22PHCS004A;
```

OUTPUT: TABLE-4

TABLE-4

STUDENT_REG_NO	STUDENT_NAME	STUDENT_ADDRESS	STUDENT_DATE_OF_BIRTH	PROGRAM_ID	SEMESTER
S001	Madesh	123 main st	26-AUG-04	101	5
S002	Suma	456 Walnut Ave	05-JUN-99	102	5
S003	Keerthi	789 Elm St	01-JUN-21	103	4
S004	Vaishu	101 Cedar Rd	10-SEP-03	104	3
S005	Pooja	202 Ash Blvd	07-OCT-03	105	2

STUDENT_RE	STUDENT_NAME	STUDENT_ADDRESS	STUDENT_DATE_OF	PROGRAM_ID	SEMESTER
-----	-----	-----	-----	-----	-----
S001	Madesh	123 main st	26-AUG-04	101	5
S002	Suma	456 Walnut Ave	05-JUN-99	102	5

S003	Keerthi	789 Elm St	01-JUN-21	103	4
S004	Vaishu	101 Cedar Rd	10-SEP-03	104	3
S005	Pooja	202 Ash Blvd	07-OCT-03	105	2

```
CREATE TABLE DEPENDENT (  
  FACULTY_ID VARCHAR(10),  
  DEPENDENT_NAME VARCHAR(30),  
  DEPENDENT_AGE INT,  
  DEPENDENT_PHONE_NUMBER INT,  
  DEPENDENT_RELATION VARCHAR(40)  
);
```

Table created.

describe dependent;

Name	Null?	Type

FACULTY_ID		VARCHAR2(10)
DEPENDENT_NAME		VARCHAR2(30)
DEPENDENT_AGE		NUMBER(38)
DEPENDENT_PHONE_NUMBER		NUMBER(38)

DEPENDENT_RELATION VARCHAR2(40)

```
INSERT INTO DEPENDENT (FACULTY_ID, DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)
VALUES
(&NEWFACULTY_ID,&NEWDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATIO
N');
```

Enter value for newfaculty_id: 1

Enter value for newdependent_name: Suma

Enter value for newdependent_age: 22

Enter value for newdependent_phone_number: 7905643190

Enter value for newdependent_relation: Sister

```
old 1: INSERT INTO DEPENDENT (FACULTY_ID, DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)
VALUES
(&NEWFACULTY_ID,&NEWDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATIO
N')
```

```
new 1: INSERT INTO DEPENDENT (FACULTY_ID,
DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION) VALUES (1,'Suma','22','7905643190','Sister')
```

1 row created.

SQL> /

Enter value for newfaculty_id: 2

Enter value for newdependent_name: siddha

Enter value for newdependent_age: 45

Enter value for newdependent_phone_number: 9034679021

Enter value for newdependent_relation: uncle

```
old 1: INSERT INTO DEPENDENT (FACULTY_ID, DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)
VALUES
(&NEWFACULTY_ID,&NEWDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATIO
N')
```

```
new 1: INSERT INTO DEPENDENT (FACULTY_ID,
DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION) VALUES (2,'siddha','45','9034679021','uncle')
```

1 row created.

SQL> /

Enter value for newfaculty_id: 3

Enter value for newdependent_name: vineesha

Enter value for newdependent_age: 20

Enter value for newdependent_phone_number: 7805434789

Enter value for newdependent_relation: elder sister

```
old 1: INSERT INTO DEPENDENT (FACULTY_ID, DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)
VALUES
(&NEWFACULTY_ID,&NEWDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATIO
N')
```

```
new 1: INSERT INTO DEPENDENT (FACULTY_ID,
DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION) VALUES (3,'vineesha','20','7805434789','elder
sister')
```

1 row created.

```
INSERT INTO DEPENDENT
(FACULTY_ID,DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)VALUES(&NEWFACULTY_ID,'&NEWDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATION');
```

Enter value for newfaculty_id: 6

Enter value for newdependent_name: Moksha

Enter value for newdependent_age: 25

Enter value for newdependent_phone_number: 7856320367

Enter value for newdependent_relation: mother in law

```
old 1: INSERT INTO DEPENDENT
(FACULTY_ID,DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)VALUES(&NEWFACULTY_ID,'&NEWDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATION')
```

```
new 1: INSERT INTO DEPENDENT
(FACULTY_ID,DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)VALUES(6,'Moksha','25','7856320367','mother in law')
```

```
select * from dependent;
```

OUTPUT: TABLE-5

TABLE-5

FACULTY_ID	DEPENDENT_NAME	DEPENDENT_AGE	DEPENDENT_PHONE_NUMBER	DEPENDENT_RELATION
6	Moksha	25	7856320367	mother in law
1	Suma	22	7905643190	Sister
2	siddha	45	9034679021	uncle
3	vineesha	20	7805434789	elder sister

FACULTY_ID DEPENDENT_NAME DEPENDENT_AGE DEPENDENT_PHONE_NUMBER DEPENDENT_RELATION

6	Moksha	25	7856320367	mother in law
1	Suma	22	7905643190	Sister
2	siddha	45	9034679021	uncle
3	vineesha	20	7805434789	elder sister

SELECT table_name FROM user_tables;

TABLE_NAME

STUDENT_DETAIL
STUDENT_4K
FACULTY
DEPARTMENT
PROGRAM
DEPT_4K
FACULTY_4K
PROGRAM_4K
JNC_DEPTS
FACULTY_22PHCS004A
DEPARTMENT_22PHCS004A

TABLE_NAME

PROGRAM_22PHCS004A
STUDENT_22PHCS004A
COURSE
STUDENT
COURSES
STUDENTS

SUBJECT
DEPENDENT
JNC_EMPLOYEE

20 rows selected.

To write SQL queries to implement the following concepts:

- 1) Select queries with WHERE clause
[A] SELECT * FROM FACULTY_22PHCS004A WHERE faculty_date_of_joining > TO_DATE('2020-01-01', 'YYYY-MM-DD');

OUTPUT: TABLE-6

TABLE-6

FACULTY_ID	FACULTY_NAME	FACULTY_ADDRESS	FACULTY_DATE_OF_JOINING	DEPT_ID
1	Kushi	123 Main St	15-JAN-20	10
4	Jo	456 Main St	18-FEB-21	40
5	Anav	567 Main St	15-JAN-20	50

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID

1 Kushi
123 Main St
15-JAN-20 10

4 Jo
456 Main St
18-FEB-21 40

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID

5 Anav
567 Main St
15-JAN-20 50

**[A-1] SELECT FACULTY_ID, FACULTY_NAME, FACULTY_ADDRESS, FACULTY_DATE_OF_JOINING FROM FACULTY_22PHCS004A WHERE
DEPT_ID='10';**

OUTPUT: TABLE-7

TABLE-7

FACULTY_ID	FACULTY_NAME	FACULTY_ADDRESS	FACULTY_DATE_OF_JOINING
1	Kushi	123 Main St	15-JAN-20

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D

1 Kushi

123 Main St

15-JAN-20

[B] SELECT * FROM DEPARTMENT_22PHCS004A WHERE DEPT_ID > 30;

OUTPUT: TABLE-8

TABLE-8

DEPT_ID	DEPT_NAME	HOD_ID
40	Physics	4
50	Maths	5

DEPT_ID

DEPT_NAME

HOD_ID

40

Physics

4

50

Maths

5

[B-1] SELECT DEPT_NAME, HOD_ID FROM DEPARTMENT_22PHCS004A WHERE DEPT_ID=50;

OUTPUT: TABLE-9

TABLE-9

DEPT_NAME	HOD_ID
Maths	5

DEPT_NAME

HOD_ID

Maths

5

[C] SELECT * FROM STUDENT_22PHCS004A WHERE STUDENT_REG_NO < 'S002';

OUTPUT: TABLE-10

TABLE-10

STUDENT_REG_NO	STUDENT_NAME	STUDENT_ADDRESS	STUDENT_DATE_OF_BIRTH	PROGRAM_ID	SEMESTER
S001	Madesh	123 main st	26-AUG-04	101	5

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

S001 Madesh

123 main st

26-AUG-04 101 5

[C-1] SELECT STUDENT_NAME FROM STUDENT_22PHCS004A WHERE SEMESTER > 4;

OUTPUT: TABLE-11

TABLE-11

STUDENT_NAME
Madesh
Suma

STUDENT_NAME

Madesh
Suma

2) Select queries with sort using order by clause
[A] SELECT * FROM FACULTY_22PHCS004A ORDER BY FACULTY_NAME ASC;

OUTPUT: TABLE-12

TABLE-12

FACULTY_ID	FACULTY_NAME	FACULTY_ADDRESS	FACULTY_DATE_OF_JOINING	DEPT_ID
2	Alya	234 Main St	10-AUG-19	20
5	Anav	567 Main St	15-JAN-20	50
3	Arav	345 Main St	22-MAY-18	30
4	Jo	456 Main St	18-FEB-21	40
1	Kushi	123 Main St	15-JAN-20	10

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID

2 Alya

234 Main St

10-AUG-19 20

5 Anav
567 Main St
15-JAN-20 50

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID

3 Arav
345 Main St
22-MAY-18 30

4 Jo
456 Main St

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID

18-FEB-21 40

1 Kushi

123 Main St

15-JAN-20 10

[A-1] SELECT * FROM FACULTY_22PHCS004A ORDER BY FACULTY_NAME DESC;

OUTPUT: TABLE-13

TABLE-13

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID

1 Kushi

123 Main St

15-JAN-20 10

4 Jo

456 Main St

18-FEB-21 40

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID

3 Arav

345 Main St

22-MAY-18 30

5 Anav

567 Main St

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID

15-JAN-20 50

2 Alya

234 Main St

10-AUG-19 20

[B] SELECT * FROM STUDENT_22PHCS004A ORDER BY STUDENT_NAME ASC;

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

S003 Keerthi

789 Elm St

01-JUN-21 103 4

S001 Madesh

123 main st

26-AUG-04 101 5

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

S005 Pooja

202 Ash Blvd

07-OCT-03 105 2

S002 Suma

456 Walnut Ave

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

05-JUN-99 102 5

S004 Vaishu

101 Cedar Rd

10-SEP-03 104 3

[B-1] SELECT * FROM STUDENT_22PHCS004A ORDER BY STUDENT_NAME DESC;

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

S003 Keerthi

789 Elm St

01-JUN-21 103 4

S001 Madesh

123 main st

26-AUG-04 101 5

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

S005 Pooja

202 Ash Blvd

07-OCT-03 105 2

S002 Suma

456 Walnut Ave

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

05-JUN-99 102 5

S004 Vaishu

101 Cedar Rd

10-SEP-03 104 3

SQL> SELECT * FROM STUDENT_22PHCS004A ORDER BY STUDENT_NAME DESC;

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

S004 Vaishu

101 Cedar Rd

10-SEP-03 104 3

S002 Suma

456 Walnut Ave

05-JUN-99 102 5

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

S005 Pooja

202 Ash Blvd

07-OCT-03 105 2

S001 Madesh

123 main st

STUDENT_REG_NO STUDENT_NAME

STUDENT_ADDRESS

STUDENT_D PROGRAM_ID SEMESTER

26-AUG-04 101 5

S003 Keerthi

789 Elm St

01-JUN-21 103 4

3)To alter the structure of the tables by adding new columns

ALTER TABLE FACULTY_22PHCS004A

ADD SALARY DECIMAL (10,2);

Table altered.

DESCRIBE FACULTY_22PHCS004A;

Name Null? Type

FACULTY_ID NOT NULL VARCHAR2(10)

FACULTY_NAME	VARCHAR2(30)
FACULTY_ADDRESS	VARCHAR2(200)
FACULTY_DATE_OF_JOINING	DATE
DEPT_ID	NUMBER(38)
SALARY	NUMBER(10,2)

```
UPDATE FACULTY_22PHCS004A
SET SALARY= 80000.00
WHERE FACULTY_ID=1;
1 row updated.
```

```
UPDATE FACULTY_22PHCS004A
SET SALARY=90000.00
WHERE FACULTY_ID=2;
1 row updated.
```

```
UPDATE FACULTY_22PHCS004A
SET SALARY=50000.00
WHERE FACULTY_ID=3;
1 row updated.
```

```
UPDATE FACULTY_22PHCS004A
```

SET SALARY=90000.00
WHERE FACULTY_ID=4;
1 row updated.

UPDATE FACULTY_22PHCS004A
SET SALARY=100000.00
WHERE FACULTY_ID=5;
1 row updated.

SELECT * FROM FACULTY_22PHCS004A;

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID SALARY

1 Kushi

123 Main St

15-JAN-20 10 80000

2 Alya

234 Main St
10-AUG-19 20 90000

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID SALARY

3 Arav
345 Main St
22-MAY-18 30 50000

4 Jo
456 Main St

FACULTY_ID FACULTY_NAME

FACULTY_ADDRESS

FACULTY_D DEPT_ID SALARY

```
-----  
18-FEB-21      40      90000
```

```
5      Anav
```

```
567 Main St
```

```
15-JAN-20      50      100000
```

```
set linesize 150
```

```
COLUMN faculty_id FORMAT A10
```

```
COLUMN faculty_name FORMAT A20
```

```
COLUMN faculty_address FORMAT A30
```

```
COLUMN faculty_date_of_joining FORMAT A25
```

```
COLUMN dept_id FORMAT 99999
```

```
COLUMN salary FORMAT 999999
```

```
SELECT * FROM FACULTY_22PHCS004A;
```

```
FACULTY_ID FACULTY_NAME      FACULTY_ADDRESS          FACULTY_DATE_OF_JOINING  DEPT_ID  SALARY
```

```
-----  
1      Kushi      123 Main St      15-JAN-20      10  80000
```

2	Alya	234 Main St	10-AUG-19	20 90000
3	Arav	345 Main St	22-MAY-18	30 50000
4	Jo	456 Main St	18-FEB-21	40 90000
5	Anav	567 Main St	15-JAN-20	50 100000

4) Select queries with sub queries

```

SELECT STUDENT_NAME, STUDENT_ADDRESS
FROM STUDENT_22PHCS004A
WHERE PROGRAM_ID IN (
    SELECT PROGRAM_ID
    FROM PROGRAM_22PHCS004A
    WHERE DEPT_ID = (
        SELECT DEPT_ID
        FROM FACULTY_22PHCS004A
        WHERE FACULTY_NAME = 'Kushi'
    )
);

```

STUDENT_NAME

STUDENT_ADDRESS

Madesh
123 main st

5)Select queries with built-in numerical functions

```
[A] SELECT AVG(salary) AS average_salary FROM FACULTY_22PHCS004A;
AVERAGE_SALARY
-----
82000
```

```
[A-1]SELECT
SUM(salary) AS total_salary,
AVG(salary) AS average_salary,
MIN(salary) AS minimum_salary,
MAX(salary) AS maximum_salary
FROM
FACULTY_22PHCS004A;
TOTAL_SALARY AVERAGE_SALARY MINIMUM_SALARY MAXIMUM_SALARY
-----
410000      82000      50000      100000
```

6)Select queries with built-in string functions
(UPPER CASE)

SELECT STUDENT_NAME, UPPER(student_name) AS uppercase_name FROM STUDENT_22PHCS004A;

STUDENT_NAME	UPPERCASE_NAME
--------------	----------------

Madesh	MADESH
Suma	SUMA
Keerthi	KEERTHI
Vaishu	VAISHU
Pooja	POOJA

SELECT FACULTY_NAME, UPPER(FACULTY_name) AS uppercase_name FROM FACULTY_22PHCS004A;

FACULTY_NAME	UPPERCASE_NAME
--------------	----------------

Kushi	KUSHI
Alya	ALYA
Arav	ARAV
Jo	JO
Anav	ANAV

(LOWER CASE)

SELECT STUDENT_NAME, LOWER(student_name) AS lowercase_name FROM STUDENT_22PHCS004A;

STUDENT_NAME	LOWERCASE_NAME
--------------	----------------

Madesh	madesh
Suma	suma
Keerthi	keerthi
Vaishu	vaishu
Pooja	pooja

SELECT FACULTY_NAME, LOWER(FACULTY_name) AS lowercase_name FROM FACULTY_22PHCS004A;

FACULTY_NAME	LOWERCASE_NAME
-----	-----
Kushi	kushi
Alya	alya
Arav	arav
Jo	jo
Anav	anav

**7)Select queries with build-in data functions
(for year)**

SELECT student_name, EXTRACT(YEAR FROM student_date_of_birth) AS Birth_year FROM student_22PHCS004A;

STUDENT_NAME	BIRTH_YEAR
-----	-----
Madesh	2004
Suma	1999

Keerthi	2021
Vaishu	2003
Pooja	2003

SELECT faculty_name, EXTRACT(YEAR FROM faculty_date_of_joining) AS joining_year FROM FACULTY_22PHCS004A;

FACULTY_NAME JOINING_YEAR

Kushi	2020
Alya	2019
Arav	2018
Jo	2021
Anav	2020

(for month)

SELECT student_name, EXTRACT(MONTH FROM student_date_of_birth) AS Birth_month FROM student_22PHCS004A;

STUDENT_NAME BIRTH_MONTH

Madesh	8
Suma	6
Keerthi	6
Vaishu	9
Pooja	10

SELECT faculty_name, EXTRACT(MONTH FROM faculty_date_of_joining) AS joining_month FROM FACULTY_22PHCS004A;

FACULTY_NAME JOINING_MONTH

Kushi	1
Alya	8
Arav	5
Jo	2
Anav	1

(for date)

SELECT student_name, EXTRACT(DAY FROM student_date_of_birth) AS birth_day FROM STUDENT_22PHCS004A;

STUDENT_NAME BIRTH_DAY

Madesh	26
Suma	5
Keerthi	1
Vaishu	10
Pooja	7

SELECT faculty_name, EXTRACT(DAY FROM faculty_date_of_joining) AS joining_day FROM FACULTY_22PHCS004A;

FACULTY_NAME JOINING_DAY

Kushi 15

Alya 10

Arav 22

Jo 18

Anav 15

Creating new table – JNC_EMPLOYEE

```
CREATE TABLE JNC_EMPLOYEE(
```

```
EMP_ID INT PRIMARY KEY,
```

```
EMP_NAME VARCHAR(30),
```

```
EMP_DEPT VARCHAR(30)
```

```
);
```

Table created.

```
SQL> insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (&newemp_id, '&newemp_name', '&newemp_dept');
```

Enter value for newemp_id: 1

Enter value for newemp_name: srinivas

Enter value for newemp_dept: BCA

```
old 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (&newemp_id, '&newemp_name', '&newemp_dept')
```

```
new 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (1, 'srinivas', 'BCA')
```

1 row created.

SQL> /

Enter value for newemp_id: 2

Enter value for newemp_name: roopa rani

Enter value for newemp_dept: hindi

old 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (&newemp_id, '&newemp_name', '&newemp_dept')

new 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (2, 'roopa rani', 'hindi')

1 row created.

SQL> /

Enter value for newemp_id: 3

Enter value for newemp_name: ruby peter

Enter value for newemp_dept: BCA

old 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (&newemp_id, '&newemp_name', '&newemp_dept')

new 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (3, 'ruby peter', 'BCA')

1 row created.

SQL> /

Enter value for newemp_id: 4

Enter value for newemp_name: jeevan

Enter value for newemp_dept: physics

old 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (&newemp_id, '&newemp_name', '&newemp_dept')

new 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (4, 'jeevan', 'physics')

1 row created.

SQL> /

Enter value for newemp_id: 10

Enter value for newemp_name: reema

Enter value for newemp_dept: botany

old 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (&newemp_id, '&newemp_name', '&newemp_dept')

new 1: insert into JNC_EMPLOYEE (emp_id, emp_name, emp_dept) values (10, 'reema', 'botany')

1 row created.

SQL> SELECT * FROM JNC_EMPLOYEE;

EMP_ID	EMP_NAME	EMP_DEPT
1	srinivas	BCA
2	roopa rani	hindi
3	ruby peter	BCA

4 jeevan physics

10 reema botany

SQL> COMMIT;

Commit complete.

Creating new table – JNC_DEPTS

```
CREATE TABLE JNC_DEPTS(  
    DEPT_ID INT PRIMARY KEY,  
    DEPT_NAME VARCHAR(30),  
    DEPT_LOCATION VARCHAR(30)  
);
```

Table created.

SQL> insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (&newdept_id, '&newdept_name', '&newdept_loc');

Enter value for newdept_id: 101

Enter value for newdept_name: BCA

Enter value for newdept_loc: cs lab

old 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (&newdept_id, '&newdept_name', '&newdept_loc')

new 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (101, 'BCA', 'cs lab')

1 row created.

SQL> /

Enter value for newdept_id: 102

Enter value for newdept_name: physics

Enter value for newdept_loc: physics lab

old 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (&newdept_id, '&newdept_name', '&newdept_loc')

new 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (102, 'physics', 'physics lab')

1 row created.

SQL> /

Enter value for newdept_id: 103

Enter value for newdept_name: hindi

Enter value for newdept_loc: ncc

old 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (&newdept_id, '&newdept_name', '&newdept_loc')

new 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (103, 'hindi', 'ncc')

1 row created.

SQL> /

Enter value for newdept_id: 104

Enter value for newdept_name: commerce

Enter value for newdept_loc: management block

old 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (&newdept_id, '&newdept_name', '&newdept_loc')

new 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (104, 'commerce', 'management block')

1 row created.

SQL> /

Enter value for newdept_id: 105

Enter value for newdept_name: electronics

Enter value for newdept_loc: electronics lab

old 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (&newdept_id, '&newdept_name', '&newdept_loc')

new 1: insert into JNC_DEPTS (dept_id, dept_name, dept_location) values (105, 'electronics', 'electronics lab')

1 row created.

SQL> SELECT * FROM JNC_DEPTS;

DEPT_ID	DEPT_NAME	DEPT_LOCATION
101	BCA	cs lab
102	physics	physics lab

103 hindi	ncc
104 commerce	management block
105 electronics	electronics lab

SQL> COMMIT;

Commit complete.

SQL> SELECT * FROM JNC_EMPLOYEE;

EMP_ID	EMP_NAME	EMP_DEPT
1	srinivas	BCA
2	roopa rani	hindi
3	ruby peter	BCA
4	jeevan	physics
10	reema	botany

SQL> COMMIT;

Commit complete.

Create new table - dependent

```
CREATE TABLE DEPENDENT (  
  FACULTY_ID VARCHAR(10),  
  DEPENDENT_NAME VARCHAR(30),  
  DEPENDENT_AGE INT,  
  DEPENDENT_PHONE_NUMBER INT,  
  DEPENDENT_RELATION VARCHAR(40)  
);
```

Table created.

describe dependent;

Name	Null?	Type

FACULTY_ID		VARCHAR2(10)
DEPENDENT_NAME		VARCHAR2(30)
DEPENDENT_AGE		NUMBER(38)
DEPENDENT_PHONE_NUMBER		NUMBER(38)
DEPENDENT_RELATION		VARCHAR2(40)

```
INSERT INTO DEPENDENT (FACULTY_ID, DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)  
VALUES
```

```
(&NEWFACULTY_ID,&NEWDEPENDENT_NAME,&NEWDEPENDENT_AGE,&NEWDEPENDENT_PHONE_NUMBER,&NEWDEPENDENT_RELATION');

```

Enter value for newfaculty_id: 1

Enter value for newdependent_name: Suma

Enter value for newdependent_age: 22

Enter value for newdependent_phone_number: 7905643190

Enter value for newdependent_relation: Sister

```
old 1: INSERT INTO DEPENDENT (FACULTY_ID, DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)
VALUES
(&NEWFACULTY_ID,&NEWDEPENDENT_NAME,&NEWDEPENDENT_AGE,&NEWDEPENDENT_PHONE_NUMBER,&NEWDEPENDENT_RELATION');

```

```
new 1: INSERT INTO DEPENDENT (FACULTY_ID,
DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION) VALUES (1,'Suma','22','7905643190','Sister')

```

1 row created.

SQL> /

Enter value for newfaculty_id: 2

Enter value for newdependent_name: siddha

Enter value for newdependent_age: 45

Enter value for newdependent_phone_number: 9034679021

Enter value for newdependent_relation: uncle

```
old 1: INSERT INTO DEPENDENT (FACULTY_ID, DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)
VALUES

```

```
(&NEWFACULTY_ID,'&NEWDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATION')
```

```
new 1: INSERT INTO DEPENDENT (FACULTY_ID,
DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION) VALUES (2,'siddha','45','9034679021','uncle')
```

1 row created.

SQL> /

Enter value for newfaculty_id: 3

Enter value for newdependent_name: vineesha

Enter value for newdependent_age: 20

Enter value for newdependent_phone_number: 7805434789

Enter value for newdependent_relation: elder sister

```
old 1: INSERT INTO DEPENDENT (FACULTY_ID, DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)
VALUES
(&NEWFACULTY_ID,'&NEWDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATION')
```

```
new 1: INSERT INTO DEPENDENT (FACULTY_ID,
DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION) VALUES (3,'vineesha','20','7805434789','elder
sister')
```

1 row created.

```
INSERT INTO DEPENDENT
(FACULTY_ID,DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)VALUES(&NEWFACULTY_ID,&NE
WDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATION');
```

Enter value for newfaculty_id: 6

Enter value for newdependent_name: Moksha

Enter value for newdependent_age: 25

Enter value for newdependent_phone_number: 7856320367

Enter value for newdependent_relation: mother in law

```
old 1: INSERT INTO DEPENDENT
(FACULTY_ID,DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)VALUES(&NEWFACULTY_ID,&NE
WDEPENDENT_NAME','&NEWDEPENDENT_AGE','&NEWDEPENDENT_PHONE_NUMBER','&NEWDEPENDENT_RELATION')
```

```
new 1: INSERT INTO DEPENDENT
(FACULTY_ID,DEPENDENT_NAME,DEPENDENT_AGE,DEPENDENT_PHONE_NUMBER,DEPENDENT_RELATION)VALUES(6,'Moksha','25','78563203
67','mother in law')
```

```
SQL> select * from dependent;
```

FACULTY_ID	DEPENDENT_NAME	DEPENDENT_AGE	DEPENDENT_PHONE_NUMBER	DEPENDENT_RELATION
6	Moksha	25	7856320367	mother in law
1	Suma	22	7905643190	Sister

2	siddha	45	9034679021	uncle
3	vineesha	20	7805434789	elder sister

FACULTY_ID DEPENDENT_NAME DEPENDENT_AGE DEPENDENT_PHONE_NUMBER DEPENDENT_RELATION

6	Moksha	25	7856320367	mother in law
1	Suma	22	7905643190	Sister
2	siddha	45	9034679021	uncle
3	vineesha	20	7805434789	elder sister

8) Select queries with inner join.

```
SQL> SELECT f.faculty_name, d.dependent_relation, d.dependent_name
2 FROM FACULTY_22PHCS004A f
3 INNER JOIN DEPENDENT d ON f.faculty_id = d.faculty_id;
```

FACULTY_NAME DEPENDENT_RELATION DEPENDENT_NAME

Kushi	Sister	Suma
Alya	uncle	siddha
Arav	elder sister	vineesha

SQL> COMMIT;

Commit complete.

9) Left join (faculty_22PHCS004A) table

```
SQL> SELECT f.faculty_name, d.dependent_name
2 FROM Faculty_22PHCS004A f
3 LEFT JOIN DEPENDENT d ON f.faculty_id = d.faculty_id;
```

FACULTY_NAME	DEPENDENT_NAME
--------------	----------------

Kushi	Suma
Alya	siddha
Arav	vineesha
Jo	
Anav	

10) Right join

```
SQL> SELECT f.faculty_name, d.dependent_name
2 FROM Faculty_22PHCS004A f
3 RIGHT JOIN DEPENDENT d ON f.faculty_id = d.faculty_id;
```

FACULTY_NAME	DEPENDENT_NAME
	Moksha
Kushi	Suma
Alya	siddha
Arav	vineesha

```
11) Full join
SQL> SELECT faculty_22PHCS004A.faculty_name, faculty_22PHCS004A.faculty_id, faculty_22PHCS004A.dept_id,
DEPENDENT.dependent_name, DEPENDENT.dependent_relation FROM faculty_22PHCS004A
2 FULL JOIN DEPENDENT ON faculty_22PHCS004A.faculty_id = DEPENDENT.faculty_id;
```

FACULTY_NAME	FACULTY_ID	DEPT_ID	DEPENDENT_NAME	DEPENDENT_RELATION
			Moksha	mother in law
Kushi	1	10	Suma	Sister
Alya	2	20	siddha	uncle
Arav	3	30	vineesha	elder sister

Jo	4	40
Anav	5	50

6 rows selected.

12) Select queries with self-join. (student table)

```
SQL> SELECT A.student_name AS student_name1, B.student_name AS student_name2, A.semester
2 FROM student_22PHCS004A A, student_22PHCS004A B
3 WHERE A.semester = B.semester
4 ORDER BY A.semester;
```

STUDENT_NAME1	STUDENT_NAME2	SEMESTER

Pooja	Pooja	2
Vaishu	Vaishu	3
Keerthi	Keerthi	4
Suma	Suma	5
Madesh	Madesh	5
Suma	Madesh	5
Madesh	Suma	5

```
SELECT A.student_NAME AS student_name1, B.student_NAME AS student_name2, A.semester
FROM student_22PHCS004A A, student_22PHCS004A B
WHERE A.student_name = B.student_name
ORDER BY A.semester;
```

STUDENT_NAME1	STUDENT_NAME2	SEMESTER
-----	-----	-----
Pooja	Pooja	2
Vaishu	Vaishu	3
Keerthi	Keerthi	4
Madesh	Madesh	5
Suma	Suma	5

```
SQL> SELECT A.student_NAME AS student_name1, B.student_NAME AS student_name2, A.semester
2 FROM student_22PHCS004A A, student_22PHCS004A B
3 WHERE A.student_reg_no <> B.student_reg_no
4 AND A.semester = B.semester
5 ORDER BY A.semester;
```

STUDENT_NAME1	STUDENT_NAME2	SEMESTER
-----	-----	-----
Madesh	Suma	5

Suma

Madesh

5